LinkedIn Content Automation Using n8n – Complete Workflow Guide

This guide explains how to build a complete **LinkedIn content automation system** using **n8n**, **Google Gemini**, **Tavily**, and **Hugging Face** – fully free, no coding required.

🟢 Node 1 – On Form Submission

Purpose:
Starts the workflow whenever a new form is submitted. The user provides a LinkedIn post topic and target audience.

Setup Summary:

- **Trigger Type:** Form Submission
- **Form Fields:**
    - Post Topic (Text)
    - Target Audience (Text)
- **Activation:** Test using the Test URL, then switch to Production URL for live use.

Example Form URL:

https://n8n.seeken.org/form-test/5be0fb7d-14a5-468a-9755-9f8ea0c7cd5f

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🧠 Node 2 – Content Researcher (AI Agent)

Purpose:
This AI agent researches topics and creates professional, engaging LinkedIn posts using real-time data from the internet. It connects to **Google Gemini** for writing and **Tavily API** for factual research.

Setup Summary:

- **Source for Prompt:** Form Input (Post Topic + Target Audience)
- **Connected Models:**
    - Google Gemini (AI Studio)
    - Tavily API (via HTTP Request)
- **Output:** Final LinkedIn post content

System Prompt:

Overview
You are an AI agent specialized in creating professional, educational, and engaging LinkedIn posts based on any topic provided by the user.

Objectives
Always begin by conducting a real-time search using the Tavily tool to gather the most accurate, up-to-date information on the topic. The post should be written to appeal to the provided target audience.

Based on your research, generate a well-structured LinkedIn post that:
- Starts with an engaging hook
- Professional in tone
- Clear and easy to read
- Educational and insightful
- Light on emojis (use only when highly relevant and minimal)
- Includes proper source attribution (e.g., “according to [source]”)
- Contains relevant hashtags to improve visibility
- Ends with a clear call to action (e.g., asking for thoughts, feedback, or shares)

Output Instructions
- Your ONLY output should be the final LinkedIn post text.
- Do not include explanations, notes, or anything beyond the post itself.

Example Workflow
1. Receive a topic (e.g., “The ROI of warehouse automation”)
2. Use Tavily to search and gather recent information or case studies
3. Draft a LinkedIn post using that research
4. Format it with source citations, clean structure, optional hashtags, and a call to action

Tavily API Integration:

- **Endpoint:** `POST https://api.tavily.com/search`
- **Docs:** Tavily API Reference

Example Request:

```
{
  "api_key": "YOUR_TAVILY_API_KEY",
  "query": "Latest trends in AI-driven content marketing",
  "search_depth": "advanced",
  "max_results": 3,
  "include_answer": true,
  "include_images": false
}
```

🎨 Node 3 – Image Generator (AI Agent)

Purpose:
Converts the LinkedIn post text into a visual image prompt that can be used with a text-to-image generator. The visuals make the posts more engaging and brand-friendly.

Setup Summary:

- **Model:** Google Gemini (AI Studio)
- **Output:** A marketing-style graphic description prompt

System Prompt:

You are an AI agent that transforms LinkedIn posts into visual prompt descriptions for generating graphic marketing materials.

Objective
- Analyze the LinkedIn post.
- Identify the main insight or message.
- Create a professional graphic prompt for AI image generation.

Output only the image prompt. Do not repeat the post or add explanations.

Style Guidelines:
- Use abstract shapes, icons, gradients, typography, and layout ideas.
- Avoid literal or hyperrealistic scenes.
- Keep it modern, clean, and professional.

Example:
A modern flat-style graphic showing a human brain connected to mechanical gears, representing the fusion of AI and automation. Minimalist background, soft gradients, clean sans-serif text placement space at the top.

🖼️ Node 4 – Hugging Face Image Model

Purpose:
Takes the visual prompt from the Image Generator and generates a real image using the **Hugging Face text-to-image model**.

Model Used: black-forest-labs/FLUX.1-schnell

Endpoint: `https://router.huggingface.co/nebius/v1/images/generations`

Example JavaScript API Call:

```
async function query(data) {
  const response = await fetch(
    "https://router.huggingface.co/nebius/v1/images/generations",
    {
      headers: {
        Authorization: `Bearer ${process.env.HF_TOKEN}`,
        "Content-Type": "application/json",
      },
      method: "POST",
      body: JSON.stringify(data),
    }
  );
  const result = await response.blob();
  return result;
}

query({
```

```
  response_format: "b64_json",
  prompt: "Astronaut riding a horse",
  model: "black-forest-labs/flux-schnell",
}).then((response) => {
  // Use image
});
```

Tips:

- Use Hugging Face free-tier API key.
- Keep prompts short, clear, and visual.
- Output connects directly to LinkedIn node.

💼 Node 5 – LinkedIn Post Node

Purpose:
Automatically publishes the final post (text + image) directly to LinkedIn.

Setup Summary:

- **Credential:** LinkedIn OAuth2
- **Resource:** Post
- **Operation:** Create
- **Post As:** Person or Company Page
- **Text:** `{{ $("Content Researcher").item.json.output }}`
- **Media Category:** Image
- **Input Binary Field:** `generated_image.png`

LinkedIn Setup Steps

1. Go to LinkedIn Developer Portal
2. Create a new app → Add name, logo, and (optional) LinkedIn page.

Under **Auth → OAuth 2.0 Settings**, set:

Redirect URL: https://<your-n8n-domain>/rest/oauth2-credential/callback

3.
4. Copy **Client ID** and **Client Secret** to n8n credential fields.
5. Connect and authorize your LinkedIn account.
6. Select whether to post as a personal profile or company page.

Tips:

- Test with text-only posts first.
- Ensure image field name matches your output node.
- Include hashtags inside your content researcher prompt.

⚙️ Final Workflow Summary

1. **Form Submission** → User inputs topic and target audience.
2. **Content Researcher** → AI researches and writes post using Tavily + Gemini.
3. **Image Generator** → AI creates an image prompt.
4. **Hugging Face Model** → Generates a professional image.
5. **LinkedIn Node** → Publishes post automatically.

🌐 Resources & Tools

- **n8n:** https://n8n.io
- **Google Gemini (AI Studio):** https://aistudio.google.com
- **Tavily API Docs:** https://docs.tavily.com/documentation/api-reference/endpoint/search
- **Hugging Face Models:** https://huggingface.co/models

✅ Congratulations! You now have a fully automated LinkedIn content creation and posting system powered by AI — completely free.